

THE QUESTION

Q.12 · DMG 2019

The hinge line of a fold is

- a** A line that divides the fold into symmetrical parts
- b** A line that is always parallel to fold axis
- c** **A line of maximum curvature in a folded bed**
- d** A line of minimum curvature in a folded bed



CORRECT ANSWER

A line of maximum curvature in a folded bed

FIGURES FROM THE TEXTBOOK

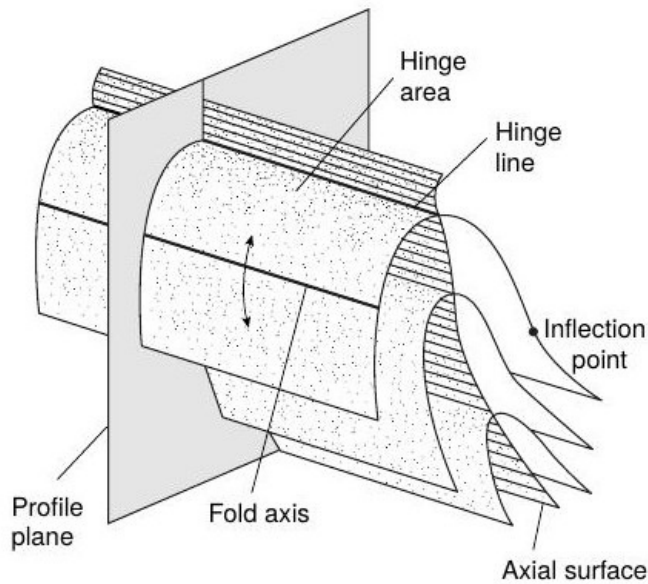


FIGURE 10.2 The terminology of a fold.

10.2 ANATOMY OF A FOLDED SURFACE 239

FIGURE 10.2 — The terminology of a fold. Note the **Hinge line** (line of maximum curvature), **Hinge area**, **Inflection point**, **Fold axis**, **Axial surface**, and **Profile plane**.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 239

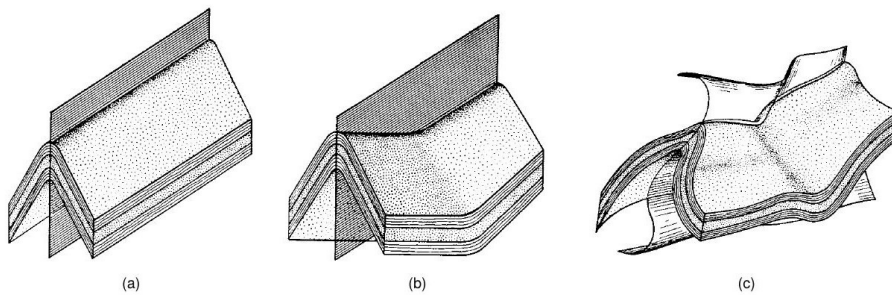


FIGURE 10.3 A cylindrical fold (a) is characterized by a straight hinge line and a noncylindrical fold (b) by a curved hinge line. The axial surface may be planar, as in (a) and (b), or curved (c).

surface can wrap partway around a cylinder. If this is

L_{∞}

FIGURE 10.3 — (a) A cylindrical fold with a **straight hinge line** parallel to the fold axis. (b) A noncylindrical fold with a **curved hinge line**. (c) A fold with a curved axial surface. This figure directly explains why option (b) is wrong: the hinge line is **NOT** always parallel to the fold axis.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 240

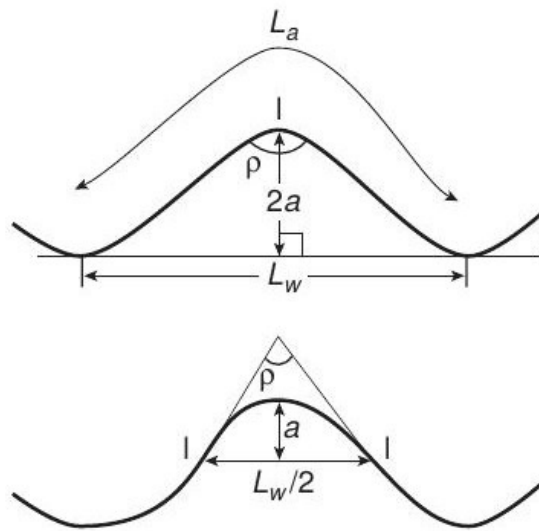


FIGURE 10.4 The interlimb angle (ρ), the wavelength (L_w), the amplitude (a), and the arc length (L_a) of a fold system in profile.

the cross-sectional plane, which has implication for the

FIGURE 10.4 — The interlimb angle (ρ), wavelength (L_w), amplitude (a), and arc length (L_a) of a fold system in profile.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 240

TEXTBOOK DEFINITIONS

VAN DER PLUIJM & MARSHAK · §10.2 ANATOMY OF A FOLDED SURFACE · PAGE 239

"The hinge area is the region of greatest curvature and separates the two limbs. **The line of greatest curvature in a folded surface is called the hinge line.** You may think of a limb as the less curved portion of a fold. In a limb there is a point where the sense of curvature changes, called the **inflection point.**"

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TABLE 10.1 — VOCABULARY OF A FOLD · PAGE 241

Hinge line: "The line of greatest curvature."

Inflection point: "The position in a limb where the sense of curvature changes."

Axial surface: "The surface containing the hinge lines from consecutive folded surfaces."

Fold axis: "Fold generator in cylindrical folds."

📖 Table 10.1 · Page 241

ON HINGE LINE VS. FOLD AXIS — PAGE 240, FOOTNOTE 1

"In the case of cylindrical folds the fold axis is of course parallel to the hinge line.¹"

¹ "Sometimes fold axis is used as a synonym for hinge line, but this is **not correct.**"

📖 Page 240

✓ Hinge Line

The line of **greatest (maximum) curvature** in a folded surface. Connects all hinge points. Visible in Figure 10.2 running along the crest of the fold.

Hinge Area

The **region** of greatest curvature separating the two limbs. The hinge line is the mathematical line of maximum curvature within this zone.

Inflection Point / Line

The point where **sense of curvature changes** (concave → convex). Located on the limbs. Represents *minimum/zero* curvature — the exact opposite of the hinge line.

Axial Surface

Surface **containing hinge lines** of all successive folded layers. Not necessarily planar (Fig. 10.3c). Does *not* necessarily divide fold symmetrically.

Fold Axis

Theoretical **generator line** in cylindrical folds only. When moved parallel to itself, sweeps out the folded surface. Parallel to hinge line *only* in cylindrical folds.

Profile Plane

The plane **perpendicular to the hinge line**. The reference plane for describing fold shape in cross-section.

Limb

The **less curved portion** of a fold, between the hinge zone and inflection point.

Interlimb Angle (ρ)

Angle between limbs measured in the profile plane. Determines fold tightness: Isoclinal (0–10°) → Tight (10–60°) → Open (60–120°) → Gentle (120–180°).

CRITICAL DISTINCTION — HINGE LINE VS. FOLD AXIS (FIGURE 10.3)

FEATURE	HINGE LINE	FOLD AXIS
Nature	Real geometric line on folded surface	Theoretical generator line (conceptual)
Definition	Line of maximum curvature	Line that, moved parallel to itself, generates the folded surface
Fold types	Exists in ALL folds	Defined only for <i>cylindrical</i> folds
Always parallel?	Only in cylindrical folds. In noncylindrical folds, hinge line curves → option (b) "always parallel" is WRONG.	

OPTION-BY-OPTION ANALYSIS

a

✗ INCORRECT

"A line that divides the fold into symmetrical parts"

This describes the **axial surface**, not the hinge line. Moreover the textbook explicitly states the axial surface "does not necessarily divide the fold into equal halves that are mirror images of one another" (p. 240). So this is doubly wrong — wrong structure named, and the description itself is not even always true.

→ What this refers to: **Axial Surface** (loosely, for symmetric folds only)

b

✗ INCORRECT — CLASSIC TRAP

"A line that is always parallel to fold axis"

True *only* for cylindrical folds. In noncylindrical folds (Figure 10.3b), the hinge line **curves** — so the fold axis concept does not even apply as a single line. The textbook's own footnote calls this confusion "not correct." The word **"always"** makes this option definitively wrong.

→ True only for: **Cylindrical folds** — not universally

c

✓ CORRECT ANSWER

"A line of maximum curvature in a folded bed"

Exact match to the textbook: "*The line of greatest curvature in a folded surface is called the hinge line*" (p. 239). Table 10.1 (p. 241) gives the same definition: **Hinge line** = **"The line of greatest curvature."** Visible in Figure 10.2 running along the apex of the folded layer.

d

✗ INCORRECT — EXACT OPPOSITE

"A line of minimum curvature in a folded bed"

This is the **inflection point/line** — where the sense of curvature reverses, located on the *limbs* of the fold. Table 10.1 (p. 241): "*Inflection point: The position in a limb where the sense of curvature changes.*" This is the geometric opposite of the hinge line.

→ What this refers to: **Inflection Point / Inflection Line**

🧠 EXAM MEMORY AID

Hinge line = Maximum curvature = Tightest bend (crest/trough zone)

Inflection line = Minimum/zero curvature = Curvature reversal on limb

Axial surface = Contains ALL hinge lines from successive layers

Fold axis || Hinge line → ONLY in *cylindrical* folds, NOT always

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The inflection point in a folded layer is defined as

- (a) The point of maximum curvature
- (b) The highest topographic point of the fold
- (c) The position where the sense of curvature changes
- (d) A point lying on the axial surface

✓ **Answer: (c) — Table 10.1, p. 241**

PROBABLE Q · 2

The axial surface of a fold is best defined as

- (a) A plane that always bisects the fold into equal mirror-image halves
- (b) The surface containing hinge lines of successive folded layers
- (c) Always a vertical plane through the fold
- (d) The plane perpendicular to the hinge line

✓ **Answer: (b) — p. 240 & Table 10.1, p. 241**

PROBABLE Q · 3

In a cylindrical fold, the fold axis is

- (a) Perpendicular to the hinge line
- (b) Contained within the limb of the fold
- (c) Parallel to the hinge line
- (d) Always horizontal

✓ **Answer: (c) — p. 240**

PROBABLE Q · 4

A fold with an interlimb angle of 45° is classified as

- (a) Gentle (120° – 180°)
- (b) Open (60° – 120°)
- (c) Tight (10° – 60°)
- (d) Isoclinal (0° – 10°)

✓ **Answer: (c) — Table 10.3, p. 246**

PROBABLE Q · 5

The profile plane of a fold is defined as the plane that is

- (a) Parallel to the hinge line
- (b) Perpendicular to the hinge line
- (c) Any vertical section through the fold
- (d) Parallel to the axial surface

✓ **Answer: (b) — p. 239–240**

PROBABLE Q · 6

Which of the following statements about a noncylindrical fold is correct?

- (a) It has a straight hinge line parallel to the fold axis
- (b) Its folded surface wraps around a cylinder
- (c) It has a curved hinge line
- (d) Its axial surface is always planar

✓ **Answer: (c) — p. 239–240 & Figure 10.3b**